

Smooth Spectral Approximation of Compact Trace Measures for Uniformly Elliptic Energy Densities

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Status

Candidate partial result, likely valid. This packet answers the smooth-measure approximation question in Vinokurov's Remark 1.3 when the energy density is continuous and uniformly positive. It does not answer the rough or degenerate weighted case.

Source Question

Vinokurov asks in Remark 1.3 of arXiv:2601.17896v3, PDF page 4, whether the eigenvalues of an arbitrary measure inducing a compact map $H^1(M, \alpha) \rightarrow L^2(M, \mu)$ can be approximated, or at least bounded from above, using smooth measures [1]. The source passage is shown below.

Remark 1.2. Thus, all absolutely continuous maximizing densities for $\bar{\lambda}_{k,p}$ are induced by p -harmonic maps. When $p = 2$, for λ_2 in dimension $m = 2$ and for λ_1 in dimensions $2 \leq m \leq 5$, there is a complementary result due to [KS3; KS2]: in the theorem above, μ may be any Radon measure for which the identity map $H^1(M) \rightarrow L^2(\mu)$ is compact. Note that absolutely continuous measures need not be of this kind.

Remark 1.3. The theorem above also holds for $\mu \in H^{1, \frac{m}{m-1}}(M)^*$ when $m \geq 3$ (see Remark 2.37 and Proposition 4.1). Those measures induce the compact identity map $H^1(M, \alpha) \rightarrow L^2(M, \mu)$ if $\alpha \geq c > 0$. However, we do not know whether the eigenvalues associated with an arbitrary measure for which the identity map $H^1(M, \alpha) \rightarrow L^2(M, \mu)$ is compact can be approximated (or at least bounded from above) by the eigenvalues associated with smooth measures.

The packet proves convergence, not merely an upper bound, under the additional hypothesis that α is continuous and bounded away from zero.

Definitions

Let (M, g) be a closed connected Riemannian manifold of dimension $m \geq 2$. For a positive density α and a finite positive Radon measure μ , write

$$a_\alpha(u, v) = \int_M \alpha \langle du, dv \rangle_g dv_g, \quad b_\mu(u, v) = \int_M \tilde{u} \tilde{v} d\mu,$$

where tildes denote quasi-continuous representatives. The variational eigenvalues are

$$\lambda_k(\alpha, \mu) = \inf_{\substack{F \subset C^\infty(M) \\ \dim F = k+1}} \sup_{0 \neq u \in F} \frac{a_\alpha(u, u)}{b_\mu(u, u)},$$

with the usual restriction that F remain $(k+1)$ -dimensional in $L^2(\mu)$.

For a Borel set $E \subset M$, let $\text{cap}_2(E)$ be its Sobolev 2-capacity, and define the local capacity modulus

$$\omega_\sigma(r) = \sup \left\{ \frac{\sigma(E)}{\text{cap}_2(E)} : E \subset M, \text{diam}(E) \leq r, \text{cap}_2(E) > 0 \right\}.$$

Maz'ya's compactness criterion states that the trace $H^1(M) \rightarrow L^2(M, \sigma)$ is compact if and only if $\omega_\sigma(r) \rightarrow 0$ as $r \downarrow 0$; see [3, Section 11.9.1] and the manifold formulation recalled in [2, Section 2.2].

Proof Intuition

Weak-star smoothing alone has the wrong one-sided min-max behavior. The key is to smooth μ without losing the quantitative local property that makes its Sobolev trace compact. We localize μ , average each piece under small diffeomorphisms, and use the quasi-invariance of capacity to keep one uniform vanishing modulus for all smoothed measures. That modulus turns weak convergence in H^1 into strong convergence in the moving spaces $L^2(\mu_n)$. Weak-star convergence then upgrades to operator-norm convergence of the quadratic mass forms. Uniform approximation of α and compact self-adjoint perturbation theory finish the eigenvalue argument.

Main Result

Theorem 1. *Let (M, g) be closed and connected, let $\alpha \in C(M)$ satisfy $0 < c \leq \alpha \leq C < \infty$, and let $\mu \neq 0$ be a finite positive Radon measure such that*

$$H^1(M, \alpha) \longrightarrow L^2(M, \mu)$$

is compact. Then there are $\alpha_n, \rho_n \in C^\infty(M)$, with $\alpha_n > 0$, $\rho_n > 0$, such that, putting $\mu_n = \rho_n v_g$,

$$\begin{aligned} \alpha_n &\longrightarrow \alpha \quad \text{uniformly,} \\ \mu_n &\xrightarrow{*} \mu, \quad \mu_n(M) = \mu(M), \\ \sup_{\|u\|_{H^1} \leq 1} |b_{\mu_n}(u, u) - b_\mu(u, u)| &\longrightarrow 0. \end{aligned}$$

Consequently, for every fixed finite variational eigenvalue,

$$\lambda_k(\alpha_n, \mu_n) \longrightarrow \lambda_k(\alpha, \mu).$$

The corresponding normalized eigenvalues converge as well.

Proof

Uniform ellipticity makes $H^1(M, \alpha)$ and the ordinary $H^1(M)$ the same Hilbert space with equivalent norms. Hence the assumed trace is compact exactly when the ordinary Sobolev trace is compact. Maz'ya's criterion therefore gives

$$\omega_\mu(r) \longrightarrow 0. \tag{1}$$

Step 1: smooth while preserving the capacitary modulus

Choose a finite smooth partition of unity $\sum_i \psi_i = 1$. On a neighborhood of $\text{supp } \psi_i$, choose compactly supported smooth vector fields $X_{i,1}, \dots, X_{i,m}$ forming a local frame. For small $z \in \mathbb{R}^m$, let $\Phi_{i,z}$ be the composition of the time- z_j flows of these vector fields. The maps $\Phi_{i,z}$ are global diffeomorphisms, converge to the identity with z , and have uniformly bounded bi-Lipschitz constants for small z .

Let η_t be a nonnegative smooth approximate identity supported in a small ball in $\mathbb{R}^m$, and set

$$\nu_t = \sum_i \int_{\mathbb{R}^m} (\Phi_{i,z})_*(\psi_i \mu) \eta_t(z) dz. \quad (2)$$

Because $z \mapsto \Phi_{i,z}(x)$ is a local submersion on the support of ψ_i , each averaged measure in (2) has a smooth density. Thus ν_t has a nonnegative smooth density, $\nu_t(M) = \mu(M)$, and $\nu_t \xrightarrow{*} \mu$.

Sobolev capacity is quasi-invariant under a uniformly bi-Lipschitz family of diffeomorphisms. Thus there are constants A, B , independent of small t, z , such that for every set E of sufficiently small diameter,

$$\begin{aligned} (\psi_i \mu)(\Phi_{i,z}^{-1} E) &\leq \mu(\Phi_{i,z}^{-1} E) \\ &\leq \omega_\mu(A \text{diam } E) \text{cap}_2(\Phi_{i,z}^{-1} E) \\ &\leq B \omega_\mu(A \text{diam } E) \text{cap}_2(E). \end{aligned}$$

After summing and averaging,

$$\omega_{\nu_t}(r) \leq B' \omega_\mu(Ar). \quad (3)$$

To make the density strictly positive while preserving mass, replace ν_t by

$$\hat{\nu}_t = (1 - \varepsilon_t) \nu_t + \varepsilon_t \frac{\mu(M)}{v_g(M)} v_g, \quad \varepsilon_t \downarrow 0.$$

The volume measure has a vanishing capacitary modulus, so (3) and (1) remain uniform for $\hat{\nu}_t$. Choose $t_n \downarrow 0$ and put $\mu_n = \hat{\nu}_{t_n}$.

Step 2: uniform modulus gives collective compactness

We record the quantitative consequence needed below. Suppose positive measures σ_n have uniformly bounded masses and

$$\Omega(r) := \sup_n \omega_{\sigma_n}(r) \longrightarrow 0. \quad (4)$$

If $u_n \rightharpoonup 0$ in $H^1(M)$, then

$$\int_M u_n^2 d\sigma_n \longrightarrow 0. \quad (5)$$

Indeed, cover M by finitely many balls B_j of radius r , with the doubled balls having bounded overlap. For every Borel set E ,

$$(\sigma_n|_{B_j})(E) = \sigma_n(E \cap B_j) \leq \Omega(2r) \text{cap}_2(E \cap B_j) \leq \Omega(2r) \text{cap}_2(E).$$

The capacitary trace inequality therefore gives a trace norm of order $\Omega(2r)^{1/2}$ for $\sigma_n|_{B_j}$. Apply it to a cutoff times $u_n - (u_n)_{2B_j}$, and then use the Poincaré inequality, to obtain

$$\int_{B_j} |u_n - (u_n)_{2B_j}|^2 d\sigma_n \leq C \Omega(2r) \int_{2B_j} |du_n|^2 dv_g.$$

For fixed r , weak convergence implies $(u_n)_{2B_j} \rightarrow 0$ for every one of the finitely many balls. Summing yields

$$\limsup_n \int_M u_n^2 d\sigma_n \leq C\Omega(2r) \sup_n \|u_n\|_{H^1}^2.$$

Letting $r \downarrow 0$ proves (5). The same localization argument, without assuming weak convergence, also gives a uniform bound for the trace norms $H^1(M) \rightarrow L^2(M, \sigma_n)$.

By (3), this lemma applies to the constructed μ_n .

Step 3: convergence in Sobolev form norm

First, for every fixed $u \in H^1(M)$,

$$b_{\mu_n}(u, u) \longrightarrow b_\mu(u, u). \quad (6)$$

For smooth u , this is weak-star convergence. For general u , approximate it in H^1 by smooth functions and use the uniform trace bounds for μ_n , together with the bounded trace for μ .

Suppose the asserted form-norm convergence were false. After passing to a subsequence, there would be u_n in the H^1 unit ball and a fixed $\delta > 0$ such that

$$|b_{\mu_n}(u_n, u_n) - b_\mu(u_n, u_n)| \geq \delta.$$

Take a further subsequence with $u_n \rightharpoonup u$ in H^1 , and set $v_n = u_n - u$. Compactness of the limiting trace gives $b_\mu(v_n, v_n) \rightarrow 0$; collective compactness gives $b_{\mu_n}(v_n, v_n) \rightarrow 0$. The mixed terms tend to zero by Cauchy–Schwarz, while (6) handles the fixed vector u . This contradicts the choice of u_n , and proves

$$\|b_{\mu_n} - b_\mu\|_{\mathcal{B}(H^1 \times H^1)} \longrightarrow 0. \quad (7)$$

Step 4: pass to eigenvalues

Choose positive $\alpha_n \in C^\infty(M)$ with $\alpha_n \rightarrow \alpha$ uniformly. Then

$$\|a_{\alpha_n} - a_\alpha\|_{\mathcal{B}(H^1 \times H^1)} \leq \|\alpha_n - \alpha\|_\infty \longrightarrow 0.$$

The form $h = a_\alpha + b_\mu$ is an equivalent Hilbert norm on $H^1(M)$: this is the standard Poincare inequality with the nonzero continuous functional $u \mapsto \int_M u d\mu$. Put $h_n = a_{\alpha_n} + b_{\mu_n}$. Relative to h , let G_n and D_n represent h_n and b_{μ_n} , respectively. Equations (7) and the uniform convergence of α_n give

$$G_n \longrightarrow I, \quad D_n \longrightarrow D \quad \text{in operator norm,}$$

where D represents b_μ . Hence the compact positive self-adjoint operators

$$S_n = G_n^{-1/2} D_n G_n^{-1/2}$$

converge in norm to D . Their ordered eigenvalues therefore converge. If $\theta_{k,n} > 0$ is an eigenvalue of S_n , the generalized eigenvalue relation is

$$b_{\mu_n}(u, v) = \theta_{k,n} h_n(u, v), \quad \lambda_k(\alpha_n, \mu_n) = \theta_{k,n}^{-1} - 1.$$

The same identity holds at the limit, so every fixed finite variational eigenvalue converges. Since the masses are equal and the relevant norms of α_n converge, normalized eigenvalues converge as well. This proves Theorem 1. $\square$

Verification Notes

The separate file `verification.md` records the adversarial check. The main points are:

1. ordinary weak-star smoothing alone is insufficient, and the proof does not use it as though it implied spectral convergence;
2. the uniform capacity estimate follows before any spectral argument and is stable under adding a vanishing positive volume component;
3. collective compactness is proved for moving measures, which is the step needed to upgrade fixed-vector convergence to operator-norm convergence;
4. the generalized eigenproblem is conjugated to self-adjoint compact operators on one fixed Hilbert space before perturbation theory is invoked.

Limitations and Novelty Check

The source question allows rough or degenerate α . The present proof uses uniform ellipticity twice: to invoke the ordinary Sobolev capacity criterion and to approximate the energy form in operator norm. For a rough weight, local diffeomorphisms need not preserve weighted capacity; for a degenerate weight, even the underlying Sobolev space can change under smoothing. Those cases remain open.

Before promotion, the run's four lightweight indexes were searched for the arXiv id and the exact spectral-approximation terms. A bounded web search used the exact question phrase, the source title/id, and combinations of compact Sobolev trace, capacity, mollification, smooth densities, and eigenvalue convergence. It found the source paper and known continuity results, but no later explicit answer or this exact approximation theorem. The main known continuity reference, Girouard–Karpukhin–Lagace, proves convergence under Sobolev-dual convergence and supplies the admissible-measure framework; it does not state the arbitrary compact-trace smoothing conclusion recorded here [2].

Human Review Recommendation

Review as a substantial partial result. The highest-value checks are the smooth localized-flow construction, the uniform capacity comparison in (3), and the localized proof of collective compactness. Do not promote the packet to a full answer without an argument for rough or degenerate α .

References

- [1] D. Vinokurov, *Eigenvalue optimization in higher dimensions and p -harmonic maps*, arXiv:2601.17896v3 (2026).
- [2] A. Girouard, M. Karpukhin, and J. Lagace, *Continuity of eigenvalues and shape optimisation for Laplace and Steklov problems*, Geom. Funct. Anal. **31** (2021), 513–561; arXiv:2004.10784.
- [3] V. G. Maz'ya, *Sobolev Spaces with Applications to Elliptic Partial Differential Equations*, Grundlehren der mathematischen Wissenschaften 342, augmented edition, Springer, 2011.